

Methods Map

STAT 109: Introductory Biostatistics

Methods Table

This page is our statistics “toolbox.” It is organized by:

- what we want to do (descriptive: plot/number summary, or inferential: test/CI/prediction), and
- what we have (the number and type of variables).

For the table below:

- columns = what we have (number/type of variables),
- rows = what we want to do (describe, infer a parameter, infer an individual value).

One toolbox table

What we want	One categorical	One numerical	Two categorical (2+ categories allowed)	EV categorical + RV numerical (2 EV groups)	EV categorical + RV numerical (3+ EV groups)	Two numerical	RV binary + EV predictor(s)
Describe: number summary	Counts, proportions, percentages (count, mutate)	Mean/median; SD/range/IQR (summarize)	Two-way counts + row/col-umn/grand proportions	Group-wise means/SDs by category (group_by)	Group-wise means/SDs by category (group_by)	Correlation <code>cor(x,y)</code> + slope/intercept summaries	Class counts and proportions by predictor groups
Describe: plot	Bar plot (geom_bar or geom_col)	Histogram / boxplot / dotplot	Stacked or 100% stacked bar (geom_col(position = "fill"))	Boxplot/histogram/dotplot per position	Boxplot per group (ANOVA context)	Scatterplot (geom_point)	Bar plots by class or probability plot vs predictor
Infer parameter: notation	p	μ	Association (χ^2 independence); 2x2: $p_1 - p_2$	$\mu_1 - \mu_2$ (numerical RV) or $p_1 - p_2$ (binary RV)	$\mu_1, \dots, \mu_k$ (ANOVA across EV groups)	β_0, β_1	Logistic β ; prob. RV given EV
Infer parameter: test name + sample null	One-proportion test, $H_0 : p = p_0$	One-sample (or paired) t -test, $H_0 : \mu = \mu_0$	Chi-square independence, H_0 : EV and RV independent	Two-sample t -test $H_0 : \mu_1 - \mu_2 = 0$, or two-proportion $H_0 : p_1 - p_2 = 0$	One-way ANOVA $H_0 : \mu_1 = \mu_2 = \dots = \mu_k$	Slope test $H_0 : \beta_1 = 0$	Logistic slope $H_0 : \beta_1 = 0$ (or relevant coefficient = 0)

What we want	One categorical	One numerical	Two categorical (2+ categories allowed)	EV categorical + RV numerical (2 EV groups)	EV categorical + RV numerical (3+ EV groups)	Two numerical	RV binary + EV predictor(s)
Infer parameter: test R code	<code>binom.test()</code> (exact) or <code>prop.test()</code>	<code>t.test(x)</code> or paired <code>t.test(x,y,paired=TRUE)</code>	<code>chisq.test(tab)</code> (paired) or <code>mcnemar.test(tab)</code> (twice):	<code>chisq.test(y ~ group, var.equal = FALSE)</code> or <code>t.test(...,paired=TRUE)</code> / <code>mcnemar.test(...)</code>	<code>fit <- lm(y ~ group); anova(fit)</code>	<code>fit <- lm(y ~ x); summary(fit)</code>	<code>fit <- glm(y ~ x_or_group, family = binomial); summary(fit);</code> compare groups with <code>anova(fit, test = "Chisq")</code>
Infer parameter: test conditions	Binomial / SRS; for z approx need $np \geq 10$ and $n(1-p) \geq 10$	Random sample (or paired) + large n or roughly normal population/differences	Independent obs.; expected counts typically ≥ 5	Two samples or random assignment; conditions for chosen method	Independent obs.; roughly normal within groups; similar spread	Linear form; independent obs.; residual assumptions for t -based inference	Binary RV; independent obs.; appropriate model form
Infer parameter: CI name	One-proportion CI	One-sample (or paired) t -interval	Context-dependent (cell/row/column or follow-up)	Two-sample t -interval or two-proportion CI	Post-hoc / group-mean intervals after ANOVA	CI for β_1 ; CI for mean response at x_0	CI for logistic β
Infer parameter: CI R code	<code>prop.test()</code>	<code>t.test()</code>	(Context-dependent; often not a single omnibus CI)	<code>t.test(...)</code> or <code>prop.test(...)</code>	<code>fit <- lm(...); then follow-up contrasts/intervals as needed</code>	<code>confint(fit, confint(fit, "x"); predict(fit, newdata=..., interval="confidence")</code>	
Infer parameter: CI conditions	Same as proportion-test approximation	Same as t -test conditions	Depends on interval used; independence remains key	Same as chosen two-group method	Same assumptions as ANOVA model	Same regression assumptions as above	Same logistic-model assumptions as above
Infer individual value: notation	Individual category outcome (usually not main target)	Individual future RV	Individual future cell/category (usually not main target)	Individual future RV in a selected EV group	Individual future RV in selected EV group level	Individual future RV at x_0	Individual $P(RV = 1 EV)$ and class after cutoff

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Infer individual value: method name	(Usually classify/report probability)	Prediction interval for one value	(Usually classify/report probability)	EV-group-based prediction interval for one value	EV-group-based prediction interval for one value	Linear-model prediction interval	Logistic probability prediction + cutoff classification
Infer individual value: R code	<code>prop.test/mo</code> based probability methods	<code>predict(fit, newdata=..., interval="prediction")</code>	context not individual prediction	<code>fit <- lm(y ~ group); predict(fit, newdata=..., interval="prediction")</code>	<code>fit <- lm(y ~ group); predict(fit, newdata=..., interval="prediction")</code>	<code>fit <- lm(y ~ x); predict(fit, family=binomial);</code>	<code>fit <- glm(..., predict(fit, newdata=..., type="response"))</code>
Infer individual value: conditions	Use caution; not usually central target	Linear-model assumptions; interval is for individual not mean	Not usually the main inferential target	Independent observations; prediction interval for individual not mean	Same as linear-model assumptions across groups	Linear form + residual diagnostics; independent observations	Binary outcome; independent observations; choose and justify cutoff

Notes

- **EV** = explanatory variable (sometimes called treatment/predictor, depending on context).
- **RV** = response variable (aka outcome variable).
- **Paired / matched data** (same unit measured twice, or matched pairs): use the **paired *t*-test** (`t.test(x, y, paired = TRUE)`) for a numerical outcome and **McNemar's test** (`mcnemar.test(tab)`) for a binary outcome — do **not** use the independent-samples versions, which assume the two measurements are unrelated.
- For any inferential row, always check:
 1. sample size(s),
 2. sampling method (random sample or independent/binomial trials),
 3. study design (observational vs experimental, paired vs independent),
 4. model assumptions where relevant.